

Cocktail Party Problem

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Abstract

Visual speech recognition (VSR), also known as lipreading, aims to recognize spoken content based exclusively on lip movements without relying on audio. While advances in deep learning and large audio-visual datasets have produced increasingly accurate VSR models, recent progress is largely attributed to massive, non-publicly available training datasets rather than architectural innovation. This paper presents a comprehensive approach demonstrating that careful model design is equally important as dataset scale. The proposed method incorporates: (1) prediction-based auxiliary tasks for jointly performing VSR and predicting audio/visual representations from pre-trained models; (2) time-masking as a novel temporal data augmentation strategy; and (3) systematic hyperparameter optimization. The resulting model achieves state-of-the-art performance across English, Mandarin, Spanish, Italian, French, and Portuguese—surpassing all prior methods trained on publicly available data by a large margin, and even outperforming models trained on up to $21\times$ more data. Incorporating additional data (including auto-transcribed or cross-lingual sources) further boosts performance, confirming both the power of model design and data scale.

1. Introduction

Visual Speech Recognition (VSR) is the task of automatically recognizing speech from video using only lip movements. Despite attracting significant research attention, early VSR systems were severely limited by (1) lack of large-scale transcribed audio-visual datasets and (2) reliance on hand-crafted visual features. The availability of datasets such as LRS2 and LRS3, combined with end-to-end deep learning frameworks, has transformed VSR into a high-accuracy, real-world-applicable technology.

However, the dominant trend in recent VSR research is to achieve state-of-the-art (SOTA) performance by scaling up training data—sometimes using tens of thousands of hours of non-public data. This paper challenges that paradigm by demonstrating that thoughtful model design—

without relying on proprietary data—can yield equally competitive or superior performance.

The key contributions of this work are:

- A novel VSR model that outperforms all prior methods trained on publicly available data by a wide margin across multiple languages.
- Introduction of prediction-based auxiliary tasks: the model jointly performs VSR and predicts audio and visual representations from pre-trained ASR/VSR models, providing additional supervisory signal.
- Proposal of time-masking as a temporal data augmentation for VSR, adapted from ASR, which forces the model to rely on contextual information for disambiguation.
- Systematic hyperparameter optimization that extracts further gains from the existing architecture.
- Empirical evidence that adding data—even from other languages or with auto-generated transcriptions—consistently improves performance.
- Discussion of open challenges and ethical considerations for VSR deployment.

2. Background & Related Work

2.1. Baseline VSR Model

The baseline VSR model follows Ma *et al.* (2021). It comprises: a 3D convolutional layer (receptive field: 5 frames) $\rightarrow$ 2D ResNet-18 $\rightarrow$ 12-layer Conformer $\rightarrow$ Transformer decoder. The model is trained end-to-end using a hybrid CTC/attention loss. Standard spatial augmentations (random cropping, horizontal flipping) are applied at training time.

2.2. Baseline ASR Model

The ASR model mirrors the VSR architecture, replacing the visual front-end (3D conv + 2D ResNet-18) with a 1D ResNet-18 that operates on raw audio waveforms. It also uses the hybrid CTC/attention framework with time-masking augmentation. Both models achieve SOTA on LRS2 and LRS3 at the time of publication.

2.3. Related Methods

Prior works such as TM-seq2seq, KD+CTC, CM-seq2seq, V2P, RNN-T, and ViT3D-TM/CM represent the landscape of VSR approaches, with recent models leveraging tens of thousands of hours of proprietary data (e.g., YT-31k: 31,000 h; YT-90k: 90,000 h). This work targets competitive performance using only publicly available datasets totaling $\sim 1,459$ hours.

3. Methodology

3.1. Model Architecture

The proposed model (Figure 1) extends the baseline VSR model with auxiliary prediction heads. It consists of four modules:

- **Front-end VSR Encoder:** 3D Conv ($5 \times 7 \times 7$, 64ch) $\rightarrow$ 3D Max Pool $\rightarrow$ ResNet-18 (2D) $\rightarrow$ Global Average Pooling $\rightarrow$ output: [B, 512, T].
- **Back-end Encoder:** Linear projection to 256-dim $\rightarrow$ 12-layer Conformer (self-attention + convolution) with relative positional encoding.
- **Decoder:** Transformer decoder with hybrid CTC/attention loss, predicting character sequences.
- **Auxiliary Predictors:** Two linear layers attached at the 6th Conformer layer, predicting audio and visual representations from frozen pre-trained ASR and VSR encoders respectively.

3.2. Prediction-Based Auxiliary Tasks

The auxiliary loss provides additional supervision to intermediate layers. Formally, let x_v and x_a denote visual and audio inputs, g_v and g_a be pre-trained visual/audio encoders (frozen), and f^l be the sub-network up to layer l . The predictors h_a and h_v minimize:

$$\mathcal{L}_{\text{AUX}} = \beta_a \cdot \|h_a(f^l(x_v)) - g_a^l(x_a)\|_1 + \beta_v \cdot \|h_v(f^l(x_v)) - g_v^l(x_v)\|_1 \quad (1)$$

where $\beta_a = \beta_v = 0.4$ (tuned on the LRS2 validation set). The total loss combines the primary VSR loss with the auxiliary loss:

$$\mathcal{L} = \mathcal{L}_{\text{VSR}} + \mathcal{L}_{\text{AUX}} \quad \text{where} \quad \mathcal{L}_{\text{VSR}} = \alpha \cdot \mathcal{L}_{\text{CTC}} + (1 - \alpha) \cdot \mathcal{L}_{\text{att}} \quad (2)$$

with $\alpha = 0.1$ (weight for CTC vs. attention loss). The auxiliary tasks encourage better visual representations at intermediate layers, leading to improved generalization.

3.3. Time Masking

Inspired by SpecAugment in ASR, time-masking replaces n consecutive frames with the mean sequence frame. This forces the model to rely on temporal context, improving

disambiguation of visually similar phonemes. One mask per second is used, with maximum duration 40% of frames per mask interval. This is the first systematic application of temporal masking in VSR training.

3.4. Hyperparameter Optimization

Systematic optimization was performed on the LRS2 validation set. Key findings: (1) Increasing batch size from 8 to 16 reduces WER; (2) Increasing the sequence-length threshold for batch halving from 150 to 220 frames also helps; (3) Larger beam size (40 for English) consistently improves decoding. The baseline architecture parameters were found to be already optimal.

3.5. Language Models

Six monolingual transformer-based language models were trained for 50 epochs. Each language LM was trained on combined corpora: English (LibriSpeech + LRS2 + LRS3 + TED-LIUM 3 + Common Voice; 166M chars), Mandarin (CMLR + news2016zh; 153M chars), Spanish (MTEDx + Common Voice + MLS; 192M chars), Italian (252M chars), Portuguese (85M chars), French (945M chars). During decoding, the LM score is weighted by language-specific β parameters.

3.6. Curriculum Learning

Training from scratch on LRS2/LRS3 is unstable due to very long utterances. The model is trained progressively: starting with videos ≤ 100 frames, then extending to 150, 300, 450, and 600 frames in successive stages. Each stage initializes from the previous checkpoint. This curriculum approach is essential for stable convergence on large-vocabulary datasets.

4. Datasets

All experiments use publicly available audio-visual datasets. Table 1 summarizes the key datasets used for training and evaluation.

Dataset	Language	Hours	Utterances	Trans.	Usage
LRS2	English	223	144,482	✓	Train / Test
LRS3	English	439	151,819	✓	Train / Test
LRW	English	157	$\sim 500K$	✓	Pre-training
CMLR	Mandarin	61	102,072	✓	Train / Test
CMU-MOSEAS (ES)	Spanish	16.3	8,582	✓	Train / Test
Multilingual TEDx (ES)	Spanish	71.4	44,745	✓	Train
Multilingual TEDx (IT)	Italian	45.8	26,387	✓	Train / Test
Multilingual TEDx (PT)	Portuguese	81.3	52,395	✓	Train / Test
Multilingual TEDx (FR)	French	84.9	58,809	✓	Train / Test
AVSpeech	English+	641	350,991	Auto	Extra Training

Table 1. Summary of datasets used in this study. ‘Auto’ indicates automatically generated transcriptions.

Pre-processing: The RetinaFace face detector and Face Alignment Network (FAN) were used to detect 68 facial

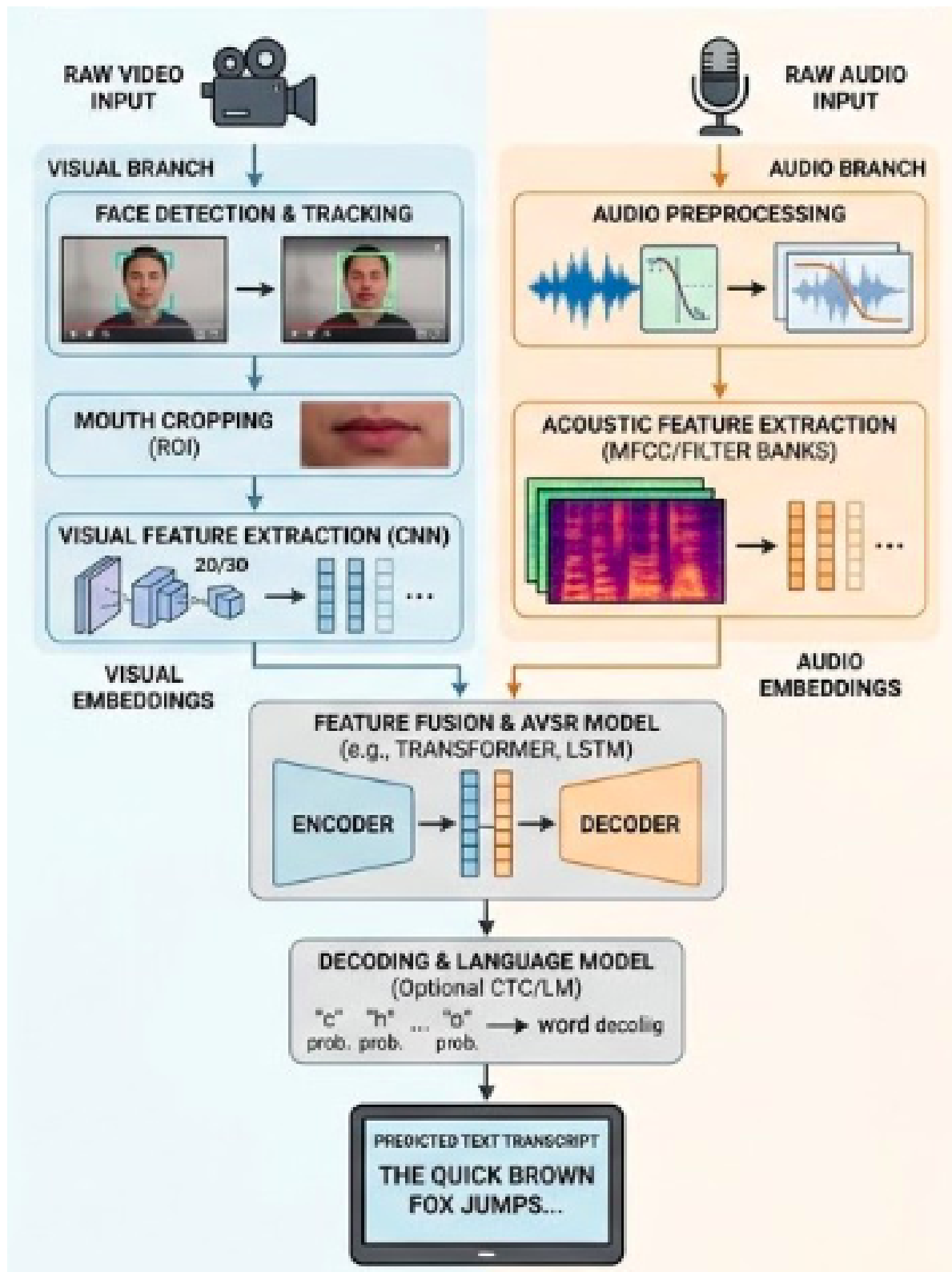


Figure 1. Overview of the proposed Visual Speech Recognition (VSR) pipeline. The model extends the baseline architecture with prediction-based auxiliary tasks, attaching linear predictors at the 6th Conformer layer to match frozen ASR and VSR representations.

landmarks. Faces were registered to a neutral reference frame via similarity transform. A 96×96 bounding box centred on the mouth was cropped, converted to grayscale, and normalized to zero mean and unit variance.

5. Experiments & Results

5.1. Evaluation Metrics

Word Error Rate (WER) is used for all languages except Mandarin, where Character Error Rate (CER) is used (since Chinese characters are not space-separated). The metric is defined as $WER = (S + D + I)/N$, where S , D , and I are substitutions, deletions, and insertions respectively, and N is the number of reference words. To produce statistically robust results, 10 independent runs are conducted per experiment and both mean $\pm$ std and best WER are reported.

5.2. Results on LRS2 (English)

The proposed method dramatically outperforms all prior methods on LRS2, achieving the best mean WER of $25.8 \pm 0.4\%$ and best single-run WER of 25.5% when using all available training data:

Method	Pre-training	Training Set	Total Hours	Mean $\pm$ Std	Best WER
MV-WAS	—	LRS2	223	—	70.4
CTC/Att.	LRW	LRS2	380	—	63.5
KD+CTC	VoxCeleb2+LRS3	LRS2	995	—	51.3
KD-seq2seq	LRW+LRS3	LRS2	818	—	49.2
TDNN	—	LRS2	223	—	48.9
CM-seq2seq	LRW	LRS2	380	—	37.9
Ours (baseline)	—	LRS2	223	33.6$\pm$0.5	32.9
Ours	LRW	LRS2	380	29.5$\pm$0.4	28.7
Ours	LRW+LRS3	LRS2	818	27.6$\pm$0.2	27.3
Ours	LRW+LRS3+AVS	LRS2	1459	25.8$\pm$0.4	25.5

Table 2. Results on LRS2. Bold rows indicate proposed method. ‘AVS’ = AVSpeech.

5.3. Results on LRS3 (English)

On LRS3, the proposed model achieves a best WER of 31.5% using all training data—outperforming models trained on up to $21 \times$ more data (RNN-T on 31,000 h scored 33.6%). Only models trained on 90,000 hours surpass this result.

Method	Total Hours	Mean $\pm$ Std	Best WER
KD+CTC	772	—	59.8
CM-seq2seq (LRW)	595	—	43.3
RNN-T (31k h private)	31,000	—	33.6
ViT3D-TM (90k private)	90,000	—	25.9
Ours (no pretraining)	438	38.6$\pm$0.4	37.9
Ours (LRW)	595	35.8$\pm$0.5	35.1
Ours (LRW+LRS2)	818	34.9$\pm$0.2	34.7
Ours (All data)	1459	32.1$\pm$0.3	31.5

Table 3. Results on LRS3. Our model outperforms all methods using publicly available data and even surpasses models trained on up to $21 \times$ more private data.

5.4. Results on CMLR (Mandarin)

The proposed model achieves a CER of 9.1% on CMLR—a remarkable 12.9% absolute improvement over the prior SOTA of 22.0% (CTCH). Cross-lingual pre-training on English data further reduces CER to 8.0%.

Method	Total Hours	Mean $\pm$ Std CER	Best CER
LipCH-Net	61	—	34.0
CSSMCM	61	—	32.5
LIBS	61	—	31.3
CTCH (prior SOTA)	61	—	22.0
Ours	61	9.1$\pm$0.05	9.1
Ours (+English pretrain)	879	8.2$\pm$0.06	8.1
Ours (+AVSpeech)	1520	8.1$\pm$0.05	8.0

Table 4. Results on CMLR (Mandarin). CER metric used. Proposed method reduces error rate by over 12 points absolute.

5.5. Results on Spanish, Italian, Portuguese, French

The model generalizes strongly across all tested languages. Key results:

- **Spanish (CMU-MOSEAS):** 7.7% absolute WER reduction over CM-seq2seq baseline; reaching 43.9% best WER with all data.
- **Spanish (TEDx):** 5.6% absolute improvement; 56.3% best WER with all data.
- **Italian (TEDx):** 5.6% improvement from model design; further 8% from additional data; best WER 57.4%.
- **Portuguese (TEDx):** 4.2% from model design; 3.9% additional gain from more data; best WER 61.5%.
- **French (TEDx):** 9.4% absolute improvement; additional 7.6% with more data; best WER 66.2%.
- **French (CMU-MOSEAS):** 11.5% absolute improvement; best WER 58.3%.

5.6. Qualitative Results

To demonstrate the real-world applicability of our model, we ran inference on a custom video clip using our end-to-end pipeline. Figure 2 shows a sample qualitative result. The model successfully tracks the facial landmarks, isolates the mouth region of interest (ROI), and decodes the visual sequence into text.

6. Ablation Study

6.1. Component Contributions

An ablation study on LRS2 and LRS3 reveals the contribution of each proposed component:

6.2. Auxiliary Loss Layer Position

The layer at which the auxiliary loss is attached was ablated across conformer layers 0–12. Layer 6 (mid-point) was found optimal on the LRS2 validation set, with both earlier and later layers yielding higher WER. This suggests



Figure 2. Qualitative inference result obtained from our pipeline-
Ground Truth: “HELLO EVERYONE I’M RECORDING THIS VIDEO FOR MY PROJECT OF SPEECH UNDERSTANDING ITS TESTING PURPOSE” **Prediction:** “HELLO EVERYONE I’M RECORDING THIS VIDEO FOR MY PROJECT OF SPEECH UNDERSTANDING UH INTERNET PURPOSE”

Configuration	WER on LRS2	WER on LRS3
Full Model (Ours)	29.5±0.4	35.8±0.5
– Audio Auxiliary Task	31.4±0.3	36.6±0.3
– Visual Auxiliary Task	30.6±0.5	36.9±0.5
– Both Auxiliary Tasks	33.2±0.5	37.8±0.6
– Time Masking	32.6±0.5	38.5±0.5
– Both Aux Tasks + Time Masking	35.0±0.5	39.1±0.4

Table 5. Ablation study on LRS2 and LRS3. Each component contributes meaningfully; removing all proposed additions approaches the CM-seq2seq baseline.

that mid-level representations are most informative for predicting audio/visual targets.

6.3. Quality of Pre-trained Models

Replacing stronger pre-trained ASR/VSR models (trained on LRW+LRS2) with weaker ones (trained on LRS2 only) consistently degrades the full model’s WER, confirming

that the quality of auxiliary targets matters. Better teacher models lead to better learned representations.

6.4. Beam Size Optimization

Beam size was optimized per language. Optimal values: English (40), Mandarin (20), Spanish (35), Italian (25), Portuguese (35), French (40). Larger beam sizes consistently reduce WER, though with diminishing returns beyond optimal values.

7. Applications, Challenges & Ethical Considerations

7.1. Applications

VSR has wide-ranging applications:

- **Audio-Visual Speech Recognition (AVSR):** VSR provides noise-robust visual stream; visual cues improve ASR by an equivalent SNR gain of ~ 15 dB.
- **Silent Speech Interfaces (SSI):** Enables communication for speech-impaired individuals (aphonia, dysphonia) by decoding mouthed speech without vocalization.
- **Video transcription:** Transcribing silent content (CCTV, archival footage, silent films) that lacks an audio track.
- **Face forgery detection:** Detecting deepfakes by identifying inconsistencies between visual lip movements and speech content.
- **Video-to-speech synthesis:** Generating audio directly from video, enabling audio inpainting and real-time speech generation.

7.2. Technical Challenges

Key open challenges include:

- **Visual ambiguity:** Different phonemes often share similar lip movement patterns (homophones), creating a fundamental performance ceiling.
- **Robustness to noise:** Sensitivity to lighting variation, occlusion, motion blur, compression artifacts, and resolution/frame-rate mismatch.
- **Pose and subject independence:** Deep networks are texture-biased, so unseen subjects or head poses can degrade accuracy.
- **Non-English and accented speech:** Most systems are designed for standard English; multilingual and accent-robust VSR remains underexplored.
- **Lombard effect and silent speech:** Performance drops when speakers exhibit hyperarticulation in noise or produce silent speech.

7.3. Ethical Considerations

VSR is a dual-use technology. Responsible deployment requires addressing:

- **Privacy and surveillance:** VSR can decode speech from CCTV or smartphone cameras without consent, posing

significant privacy risks—particularly as camera quality improves.

- **Bias and fairness:** Training data sourced predominantly from TV and TED talks likely underrepresents younger people, non-native speakers, and diverse cultural backgrounds. Demographic diversity in training data is essential for equitable performance.
- **Regulatory framework:** Proactive government regulation analogous to facial recognition—auditing before commercialization to prevent misuse. The cautionary example of unregulated facial recognition (subsequently banned in several cities) is directly applicable.
- **Positive applications:** In clinical settings (e.g., assisting tracheostomy/laryngectomy patients), VSR can be transformative. Responsible commercialization pathways already exist (e.g., Liopa’s NHS-trial application).

8. Implementation Details

All experiments were implemented using ESPNet. Training used the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$. Learning rate: linear warmup over 25,000 steps to peak 0.0004, then inverse-square-root decay. Training: 50 epochs, batch size 16, model averaging over last 10 checkpoints for inference. Pre-processing: RetinaFace + FAN for face detection/alignment; 96×96 grayscale mouth crop, normalized per training set statistics.

Architecture specifics: The Conformer back-end uses 12 layers, 4 self-attention heads projecting to 64-dim each (concatenated to 256-dim), feed-forward layers expanding to 2048-dim with ReLU + dropout ($p = 0.1$). The decoder uses standard multi-head attention with encoder-decoder cross-attention. Language model weights: $\lambda = 0.1$ (CTC weight); β varies by language {EN: 0.6, ZH: 0.3, ES: 0.4, IT: 0.5, PT: 0.3, FR: 0.3}.

9. Conclusions

This work demonstrates that state-of-the-art visual speech recognition does not require massive proprietary datasets—thoughtful model design can be equally impactful. The proposed approach combines prediction-based auxiliary tasks, time-masking augmentation, and systematic hyperparameter optimization to achieve SOTA performance across six languages using only publicly available data. Notably, the model surpasses systems trained on up to 21 times more data on English benchmarks and delivers dramatic error rate reductions on Mandarin and other languages.

The work also highlights that using more data—whether from other languages or with auto-generated transcriptions—reliably improves performance, corroborating the hypothesis that much of the recent progress in the field is driven by data scale. Future directions include improving robustness to visual noise, addressing demographic

bias in training data, developing pose-invariant architectures, and tackling underexplored speech conditions such as the Lombard effect and silent speech.

Need for proactive ethical frameworks and regulatory oversight before VSR technology is widely deployed, particularly to address privacy concerns and ensure equitable performance across diverse user populations.

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